uno: A Fast, Reproducible, Dependency-Light Rust Engine for MS1-Only Protein Identification

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Abstract

MS1-only protein identification (the DirectMS1 approach) infers peptide and protein identities from precursor mass and retention time (RT) alone, without MS2 fragmentation, enabling very short LC gradients and high sample throughput. The reference implementation, ms1searchpy, is written in Python and depends on the deep-learning RT predictor DeepLC. We present **uno**, an open-source MS1-only search engine written in Rust that consumes aligned feature matrices from the koth feature finder. uno delivers four practical advantages over existing tooling at equal-or-greater identification depth: (i) **reproducibility** — on identical input, ms1searchpy returned 152–303 proteins across repeated runs ($\sigma \approx 55$) whereas uno is deterministic ($\sigma \approx 1$), byte-identical by construction; (ii) **a distilled, pure-Rust RT model** that reproduces DeepLC's MS1 identification depth ($R^2 = 0.980$ vs DeepLC predictions) while running in-process and removing the Python/DeepLC dependency entirely; (iii) **on-data RT fine-tuning** — refitting only the pretrained RT network's linear readout over its frozen embedding on a run's own confident hits raises identification depth (+51% on the benchmark here) while remaining FDR-honest under two-proteome entrapment (zero entrapment false positives); and (iv) **speed** — algorithmic and parallel optimization of the protein-inference grid reduces an end-to-end search from 43 s to 3.4 s ($\approx 13\times$) with byte-identical output. A two-proteome entrapment harness confirms the protein-level FDR is approximately honest and, using the same harness, shows that a learned (LDA) protein rescorer inflates the empirical FDR despite raising apparent counts — underscoring why a deterministic binomial scorer is retained, and why the RT fine-tune (which passes the same test) is adopted instead.

Keywords MS1-only proteomics · DirectMS1 · retention time prediction · reproducibility · Rust · open-source software · label-free quantification

1 Introduction

Bottom-up proteomics conventionally identifies peptides from MS2 fragment spectra. MS1-only identification (DirectMS1 [1]) takes a different route: it matches the accurate precursor mass of an LC–MS feature to a theoretical peptide and confirms the assignment with predicted retention time, omitting MS2 entirely. Because no time is spent acquiring fragment spectra, MS1-only workflows support very short gradients (minutes) and correspondingly high throughput, at the cost of an information-poor identification problem: mass and RT alone are weak discriminators, and identity must be concentrated at the protein level across multiple consistent peptides.

The established open-source tool for this approach, **ms1searchpy** [2], couples a precursor mass search to RT prediction by **DeepLC** [3] and a LightGBM-based rescoring stage, with protein-level inference by a binomial model. It is effective but has two practical costs that matter at scale. First, its identification counts are **not reproducible run to run**: the DeepLC and LightGBM stages are stochastic and unseeded, so repeated searches of the same features yield materially different protein lists (§3.1). Second, it carries a **heavyweight Python/deep-learning dependency** (DeepLC, TensorFlow) whose RT-prediction step dominates wall-clock time for a single run.

We present **uno**, an MS1-only identification and label-free quantitation engine written in Rust. uno is one component of an all-Rust pipeline: the **koth** feature finder [4] performs alignment, deisotoping and XIC/LFQ; uno consumes koth's aligned feature matrix and assigns peptide/protein identities, controls FDR, and rolls intensities up into quant matrices. uno's design goals are determinism, speed, and minimal external dependencies, while meeting or exceeding the identification depth of ms1searchpy. The contributions of this work are:

1. A clean, open-source Rust reimplementation of DirectMS1-style MS1 search with **deterministic** output (§2).
2. A quantitative demonstration that **MS1-only identification counts are strongly implementation-dependent**: ms1searchpy spans 152–303 proteins on one fixed feature set, and uno sits at its median with $\approx 50\times$ lower variance (§3.1).
3. A **distilled, pure-Rust RT model** with engineered order-aware features that reproduces DeepLC's MS1 identification depth ($R^2 = 0.980$) without a Python dependency (§3.3).
4. An **on-data RT fine-tuning** step that refits only the pretrained network's readout over its frozen embedding on a run's confident hits, raising depth while passing two-proteome entrapment (§3.7).
5. **Performance engineering** of the protein-inference stage (memoized binomial, best-first greedy parsimony with lazy invalidation, parallel threshold grid) that reduces a search from 43 s to 3.4 s with byte-identical output (§3.8).

2 Experimental Procedures (Methods)

2.1 Input and pipeline position

uno consumes the aligned feature matrix produced by koth: per-feature `massCalib`, `mz`, `charge`, `rtApex`, ion mobility (`im`, where available), `combined_score`, and per-sample intensities. RT is in koth's reference-run space, so a single calibration suffices rather than per-file calibration.

2.2 Peptide database and decoys

In-silico tryptic digestion is performed with sage-core [5] (enzyme rules, missed cleavages, length/mass bounds, fixed/variable modifications), and masses are recomputed in `f64`. Candidate lookup is a binary search over a mass-sorted peptide vector. Decoys are generated by sequence reversal with an optional mass shift; we use mass-shifted decoys because they provide a clean protein-level FDR null, whereas same-mass reversed decoys are RT-indistinguishable "twins" of their targets under composition-based RT models (§3.4).

2.3 Feature–peptide matching

Each feature is matched to all peptides within a ppm tolerance, enumerating isotope-error offsets (0, ±1 Da/charge). The resulting peptide–feature matches (PFMs) are reduced to the top-K best-mass candidates per feature to bound the downstream cost without discarding the RT-correct assignment.

2.4 Retention-time model (this work, §3.3)

uno predicts RT with a model **distilled from DeepLC**: DeepLC predictions over a large, diverse tryptic digest serve as training labels. Sequences are encoded by a 99-dimensional **order-aware feature set**: amino-acid composition (20), N- and C-terminal residue identities (first three, last three; 6), **position-binned composition** (three sequence bins × 20 = 60), and 13 scalar descriptors (length, ln mass, Kyte–Doolittle [6] / Krokhin-SSRCalc [7] / Eisenberg [8] hydrophobicity sums, an N-terminus-weighted SSRCalc term, proline/aromatic/basic counts, and a hydropathy moment). The shipped student is a small multilayer perceptron over these features (a gradient-boosted [9] variant is also supported); it is distilled offline once and predicts RT for all candidate peptides in-process at search time.

Because the frozen predictor lives in DeepLC's abstract RT space, uno calibrates it to the run's observed gradient using a RANSAC-seeded, monotone piecewise-linear bootstrap on confident identifications, progressively tightened across rounds and vouched by protein-level multiplicity.

2.5 On-data RT fine-tuning (this work, §3.7)

A frozen predictor plus monotone calibration can only **rescale** the prediction axis; it cannot correct per-peptide ranking errors the network makes on an unseen gradient. Refitting a fresh linear model on raw composition features (the DirectMS1 on-data trick) discards the pretrained representation and underperforms (§3.7). Instead, uno **fine-tunes only the readout**: it freezes the MLP's learned nonlinear embedding (the penultimate-layer activations) and refits a ridge-regularized linear map from that embedding to the run's **observed** RT, using the confident-hit set with iterative outlier rejection. This keeps the pretrained representation while adapting offset, scale, and shape to the run. The fit is closed-form and deterministic, so it preserves byte-identical output. The same confident-hit selection and decoy treatment apply, so target/decoy FDR remains calibrated (validated by entrapment, §3.7).

2.6 Scoring and FDR

PFMs are scored by a deviation discriminant combining recalibrated mass error (in σ units, via sigma-clipped re-estimation of the mass-error distribution) and calibrated RT deviation. Because mass and RT alone do not separate target from decoy at the single-peptide level

in this regime (§3.2), identity is concentrated at the protein level: a binomial survival score $-\log_{10} P(X \geq k \mid n, p)$ evaluates whether a protein's k observed peptides exceed the random-match background rate p estimated from decoys, followed by greedy parsimony, target/decoy pair competition, and a target/decoy FDR walk. Mass/RT acceptance thresholds are optimized on a grid to maximize target proteins at the chosen FDR (mirroring ms1searchpy's `worker`); because decoys pass the identical filter at every grid point, the FDR estimate stays calibrated regardless of which cutoff maximizes yield.

All stages are implemented to be **deterministic**: there is no use of process-randomized iteration order or unseeded RNGs in any result-affecting path, so repeated runs on identical input produce byte-identical output (§3.1). An optional learned (cross-validated LDA) protein rescorer is provided but **disabled by default**, because entrapment validation shows it inflates the empirical FDR (§3.6); the binomial model with on-data-fine-tuned RT is the validated default.

2.7 Quantitation

Identified peptides/proteins are mapped onto koth's per-sample intensities to emit peptide- and protein-level quant matrices; uno does not re-quantify (koth owns alignment/XIC/LFQ).

2.8 Benchmark data and comparison

Variance and RT-distillation results (§3.1, §3.3, §3.4) use the HeLa dataset PXD015669 (run `QEHF1_09744_JB`, Q Exactive HF), with features called by koth. The on-data fine-tune and performance results (§3.7, §3.8) are measured on a koth-aligned benchmark feature matrix (20 samples, 45,155 aligned features) searched against a human + *E. coli* database (24,819 proteins) with the distilled MLP RT model; the entrapment experiments add a *S. cerevisiae* proteome (6,068 proteins) as a known-absent probe. ms1searchpy and uno were run on the identical koth feature set where compared.

3 Results and Discussion

3.1 MS1-only identification counts are strongly implementation-dependent

Running ms1searchpy on a single fixed koth feature set six times yielded **152, 153, 155, 188, 203, and 303 proteins** at a nominal 1% FDR ($\sigma \approx 55$; the logged random-match background p ranged 0.016–0.037). The high outlier (303) is consistent with the stochastic LightGBM rescoring occasionally overfitting the target/decoy boundary. uno, run on the same features through its deterministic path, returned **175–179 proteins** ($\sigma \approx 1$) — sitting at the **median** of ms1searchpy's distribution with $\approx 50\times$ lower variance.

uno's determinism is by design. Two sources of run-to-run nondeterminism in early development were traced to Rust's process-randomized hash-map iteration order: (i) the RANSAC RT-calibration seed was assembled from a hash map and sampled positionally, and (ii) the protein greedy-parsimony and FDR steps broke score ties by iteration order. Both were made order-independent (sort the calibration seed by feature index; resolve ties on accession). After these fixes, independent runs on identical input are **byte-identical**. Determinism is thus a guaranteed property of the implementation, not a statistical observation.

3.2 Identity is concentrated at the protein level, not the peptide level

We confirmed directly that in this regime mass+RT cannot separate target from decoy peptides: among matches with near-zero RT deviation (< 0.01 min), target and decoy peptides occur at $\approx$ 50:50. Most target peptides in a proteome are not actually present in a given sample, so a random target match is no more likely than a decoy match. Discrimination therefore arises only from **multiplicity** — real, present proteins accumulate several RT-consistent peptides while decoy proteins remain singletons — which the binomial protein model exploits. This property is intrinsic to MS1-only search and bounds what any single-peptide discriminant (including ML rescoring) can achieve.

3.3 A distilled pure-Rust RT model reaches DeepLC's identification depth

The RT predictor is the heaviest dependency in MS1-only search. A naïve composition-feature model (the prior 69-d Klammer set) trailed: $R^2 = 0.972$ vs DeepLC predictions, and 141 proteins / 119 reference-overlap on the benchmark (vs the DeepLC library's 176 / 152). Replacing the encoding with the 99-d **order-aware** feature set (§2.4) raised the distilled model to $\boldsymbol{R^2 = 0.980}$**, matching the LightGBM ceiling**, and restored identification depth to **DeepLC parity** (Table 1), achieved **in-process in pure Rust** with no Python dependency. The feature encoding, not the boosting backend, was the lever; a still-richer 215-d set gave only marginal improvement ($R^2 = 0.9824$), indicating the 99-d set is near the practical ceiling for hand-crafted features.

RT model	R^2 vs DeepLC	Proteins / ref-overlap	RT-stage cost	Dependency
Composition (69-d)	0.972	141 / 119	in-process	none (Rust)
Order-aware (99-d, this work)	**0.980**	**~180 / ~152**	**in-process**	**none (Rust)**
DeepLC library	(teacher)	176 / 152	~130 s preproc.	Python/TF

Table 1: RT model vs identification depth on the HeLa benchmark (1% FDR).

3.4 A reverse-sequence benchmark quantifies sequence-order capture

A peptide and its reverse have identical amino-acid composition but different true retention time; an order-aware predictor must assign them different RTs, while a composition-only model is mathematically forced to predict the same value for both. This forward−reverse RT difference is precisely the signal that lets a search engine distinguish a real peptide from a reversed decoy.

We constructed a benchmark of 46,304 held-out tryptic peptides and their reverses (non-tryptic, out-of-distribution), predicted DeepLC RT for both, and measured how well each model reproduced the forward−reverse RT shift. A composition-only model captured **0%** of this signal (by construction). uno's order-aware model captured **72%** (Pearson $r = 0.68$) while predicting reverse-sequence RT nearly as well as forward ($R^2_{\text{rev}} = 0.96$, $R^2_{\text{fwd}} = 0.98$; Table 2). The residual 28% is the genuine gap between hand-crafted features and DeepLC's sequence convolution. This benchmark is, to our knowledge, a novel and reusable way to report RT-model sequence-order sensitivity.

Model	R^2_{fwd}	R^2_{rev}	order-Δ corr.	order magnitude
Composition-only	0.96	0.95	0.00	0%
uno order-aware (this work)	**0.98**	**0.96**	**0.68**	**72%**

Table 2: Reverse-sequence RT benchmark (46,304 held-out peptides + reverses).

3.5 Two-proteome entrapment confirms the protein-level FDR is honest

Target/decoy FDR can be optimistic if the decoys do not faithfully model the random-coincidence behaviour of real-but-absent peptides. We tested uno's reported FDR directly with a two-proteome entrapment: the sample was searched against the real database plus an entrapment proteome — *S. cerevisiae*, absent from the sample — together with decoys for both. Any accepted yeast protein is a known false positive. We estimate the empirical false-discovery proportion as $n_{\text{yeast}} \times \frac{N_{\text{real}}}{N_{\text{yeast}}}/n_{\text{real}}$, the DB-size correction for the smaller entrapment proteome. At a reported 1% protein FDR, uno's binomial scorer is approximately honest (Table 3).

3.6 On-data RT fine-tuning raises depth while preserving FDR

We evaluated the on-data RT options of §2.5 on the koth-aligned human+*E. coli* benchmark matrix. The frozen MLP with monotone calibration identified **39 proteins** at 1% FDR. Refitting a fresh composition-feature ridge on the confident hits — the literal DirectMS1 on-data trick — **degraded** depth to **9 proteins**: a from-scratch linear model is a far weaker RT predictor (residual $\sigma \approx 0.56$ min vs the frozen model's ≈ 0.2 min) and discards the pretrained representation. Fine-tuning only the readout over the frozen MLP embedding (§2.5) instead raised depth to **59 proteins** (+51% over frozen), the configuration now shipped as default.

To distinguish real gain from FDR leakage we swept the ridge penalty: depth was stable across $\lambda = 1 \to 10$ ($59 \to 56$) and collapsed only when over-regularized into under-fitting ($\lambda = 100 \to 19$, $\lambda = 1000 \to 4$). A genuine overfit would peak sharply at low λ and crumble with any regularization; the observed plateau is the signature of real signal. We then ran the entrapment harness of §3.5 (real DB + *S. cerevisiae* probe). Both the frozen and fine-tuned configurations accepted **zero** yeast proteins at 1% FDR — the nearest yeast protein sat at $q = 0.15$, $15\times$ past threshold — while the fine-tune added real proteins ($34 \to 38$ on the entrapment DB; Table 3). The fine-tune is therefore FDR-honest: it is the mirror image of the LDA rescorer (§3.6), gaining depth **without** admitting entrapment false positives.

Scorer / RT mode	Real accepted	Yeast (false)	Empirical FDR
Binomial (default)	201	1	1.7%
LDA rescorer (naïve)	208	7	11.5%
LDA rescorer (hardened CV)	209	7	11.4%
RT fine-tune ablation (this revision)			
Frozen MLP + calibration	34	0	~0%
MLP readout fine-tune (default)	**38**	**0**	**~0%**

Table 3: Two-proteome entrapment. Binomial and LDA rows: HeLa human sample vs human + yeast (prior operating point). RT-fine-tune rows: koth human+*E. coli* benchmark vs real DB + yeast probe (this revision). "Empirical FDR" is the DB-size-corrected entrapment estimate at a reported 1%.

A limitation of the probe used here is resolution: with zero entrapment hits a single yeast acceptance would scale (×4.09 DB-size factor) to ≈ 10% estimated FDR, so "0%" means "below this test's floor," not "provably zero." The **comparative** claim — that the fine-tune adds no entrapment false positives relative to the validated-honest frozen path — is robust; tightening the absolute estimate requires a size-matched probe proteome (§5).

3.7 Learned protein rescoring raises apparent counts but inflates FDR

We separately evaluated a learned protein-level rescorer combining the binomial score with log peptide count, coverage, and RT/mass consistency, implemented as a deterministic closed-form Fisher LDA with 3-fold cross-validation by accession hash (no RNG). It raised apparent identifications (218 → 242 proteins) and was byte-identical across runs. However, the **same entrapment test exposed the gain as illusory**: the LDA accepted **7 yeast (false) proteins versus the binomial's 1**, an estimated empirical FDR of **11.5%** at a nominal 1% (Table 3).

Hardening the rescorer with textbook semi-supervised safeguards — high-confidence iterated positives ($q \leq 0.01$, refreshed), target/decoy pairs co-folded, and stronger covariance shrinkage — **removed the inflated apparent count (242 → 223) but did not restore FDR control** (still 11.4% empirical). The inflation is therefore **not classical overfitting** but an **objective mismatch**: a learned scorer is trained to separate targets from decoys, yet the false positives that matter are **real-but-absent** proteins, whose RT and mass consistency resemble present proteins and differ from decoys. The very features that could add depth are exactly those a present and an absent-but-real protein share. The count-based binomial is robust precisely because it uses only peptide multiplicity, where an absent protein and a decoy are indistinguishable. We therefore **retain the binomial scorer as default** and ship the LDA disabled [10]. This is a transferable cautionary result: learned protein rescoring in MS1-only search trades FDR integrity for apparent depth even under careful regularization, and entrapment validation is essential to detect it. The RT fine-tune (§3.7) is adopted precisely **because** it passes this same test where the LDA fails.

3.8 Performance engineering

The end-to-end search was profiled on the benchmark matrix (20 cores). The original implementation ran in **43 s**, of which **~40 s** was protein inference. The cost was the threshold-optimization grid (§2.6): ~99 (RT, mass) cells, each running a greedy parsimony loop that

re-scored every surviving protein on every round. Instrumentation showed up to **17.7 million** exact-binomial evaluations per cell (incomplete-beta continued fraction), ~1 billion across the grid.

Three changes removed this without altering output: (i) the independent grid cells were run in **parallel** (rayon), with the winner selected by the same deterministic rule; (ii) within each cell the binomial was **memoized** on $(n_{\text{theo}}, n_{\text{obs}})$, collapsing ~18 M evaluations to a few thousand distinct ones, and the greedy loop was restructured as a **best-first heap with lazy invalidation** — only the few proteins that lose a peptide when a winner is chosen are re-scored and re-pushed, turning the per-round full rescan into work proportional to peptides removed; (iii) a release profile with link-time optimization and a single codegen unit.

The protein phase fell from **~40 s to ~1 s** single-threaded (the algorithmic gain alone, $\approx 40\times$), and the full search from **43 s to 3.4 s** ($\approx 13\times$). Output is **byte-identical** to the original across the whole pipeline (identical protein lists and FDR), and remains deterministic across runs.

4 Limitations

We report these candidly, as they delimit where MS1-only search and uno apply.

- **Inherent PSM-FDR floor.** Mass+RT cannot separate target/decoy at the single-peptide level (§3.2); identification depends on protein-level multiplicity. This is a property of MS1-only data, not of uno.
- **No ion mobility, no peptide-centric depth.** On Orbitrap data without ion mobility, peptide-centric extraction from raw hills does not add identifications: the (m/z, RT) coordinate is too degenerate without a third dimension. Ion-mobility data (timsTOF) is expected to lift this; uno predicts CCS via an analogous distilled model where IM is present.
- **Search-breadth expansion is counterproductive.** Adding missed cleavages or modifications inflates the theoretical count faster than the observed count and raises the decoy background, reducing identifications — consistent with DirectMS1′s minimal digest.
- **TMT-labeled MS1 data.** uno currently yields ≈ 0 identifications on TMT MS1 data; DeepLC RT for TMT-modified peptides correlates only $r \approx 0.71$ with observed RT, too weak to seed calibration.
- **Entrapment resolution.** The single-probe entrapment bounds the empirical FDR only coarsely (§3.7); a size-matched probe and a multi-dataset panel are required to tighten absolute estimates.

5 Validation roadmap

1. **Size-matched / multi-dataset entrapment** to tighten the absolute FDR estimate for both the binomial default and the RT fine-tune.
2. **Multi-dataset benchmark** (≥ 3 datasets) spanning instruments/gradients/ organisms; per dataset: uno vs ms1searchpy protein counts, FDR validity, variance ($n \geq 10$ runs), and speed, including the fine-tune depth gain.
3. **Formal reproducibility experiment** ($n \geq 10$ runs $\times$ both engines $\times$ each dataset).
4. **Quantitation validation** (known-ratio sample, CV/accuracy) or explicit ID-only scope.
5. Software-release hygiene for open-source expectations: a build that does not require a local sage checkout, README, example data, tests/CI, and an archived release with DOI.

6 Conclusions

uno is a fast, deterministic, open-source Rust engine for MS1-only protein identification that meets or exceeds the identification depth of ms1searchpy while removing its run-to-run variability and its heavyweight Python/DeepLC dependency. Its output is byte-identical across runs by construction; its distilled order-aware RT model reaches DeepLC's MS1 depth in pure Rust; an on-data fine-tune of the RT readout adds further depth (+51% here) while passing two-proteome entrapment; and targeted performance engineering brings an end-to-end search to a few seconds with unchanged results. The accompanying reverse-sequence benchmark provides a direct, reusable measure of RT-model sequence-order sensitivity, and our entrapment analysis shows that learned protein rescoring — while raising apparent counts — inflates FDR, motivating uno's transparent binomial model and the entrapment-validated RT fine-tune adopted in its place.

7 Data and Software Availability

uno is open-source (MIT) at [URL]. Benchmark data: PXD015669. RT distillation labels, trained models, and analysis scripts: [DOI/Zenodo — TODO].

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